

Sentiment Analysis of Indonesian Political News using Indobert

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Abstract—Automated sentiment analysis of Indonesian political news faces significant challenges, including complex journalistic language and severe class imbalance. This study evaluates IndoBERT word embeddings by comparing Deep Learning architectures (Fine-Tuning, CNN, BiLSTM, GRU) against Machine Learning Ensembles (Random Forest, XGBoost) to address extreme data imbalance. Using an original dataset of 2,599 annotated news articles (dominated by a 62% Neutral class), the methodology incorporated class weight amplification and Focal Loss. Experimental results demonstrate that Deep Learning models significantly outperform traditional ensembles. The IndoBERT + GRU architecture emerged as the most stable model, achieving the highest Macro F1-Score of 82.81%. Its gating mechanisms effectively retained long-term contexts and overcame majority class bias, achieving a 71.95% Recall for the minority Positive class. In contrast, Machine Learning ensembles exhibited extreme bias, with Random Forest nearly failing to detect minority samples (10.98% Recall). These findings establish that integrating IndoBERT with GRU, optimized via fine-tuning and Focal Loss, provides a robust solution for imbalanced text classification, enabling reliable public opinion monitoring.

Keywords— *Sentiment Analysis, IndoBERT, Imbalanced Data, Deep Learning.*

I. INTRODUCTION

The evolution of mass media in Indonesia following the reform era has transformed it into a fundamental pillar of democratic dynamics. Media is no longer merely an information channel, but a powerful instrument that influences public opinion and monitors political policies [1]. Driven by rapid digitalization, online news portals now publish thousands of articles daily. This creates a massive surge in text data volume, particularly ahead of critical events such as national elections.

This overwhelming volume of news coverage has fundamentally altered how the public consumes political information. Conventional sentiment and public opinion mapping are no longer efficient and are highly susceptible to subjective bias. Relying on manual analysis to dissect

thousands of daily news texts severely bottlenecks the ability to swiftly understand shifts in public opinion.

Consequently, implementing automated sentiment analysis using NLP has become essential. Such computational approaches enable researchers, political observers, and policymakers to objectively classify the sentiment of political news. By leveraging data-driven methods, the dynamics of public opinion derived from online news can be measured with precision and speed [2], [3].

Despite its urgency, applying sentiment analysis to Indonesian-language political news presents complex computational challenges. First, journalistic language often utilizes long sentences, formal vocabulary, and implicit information delivery [4]. Second, the data is inherently imbalanced. Because journalistic texts adhere to strict standards of neutrality and ethical objectivity [5], [6], neutral sentiment heavily dominates the corpus. In this study's dataset—collected from CNBC, Detik, and Kompas—over 62% of the 2,599 samples are classified as neutral (1,614 neutral, 574 negative, and 411 positive). This severe imbalance frequently causes classification models to bias toward the majority class, thereby degrading detection performance for minority classes [7].

Previous studies have attempted to implement NLP in Indonesian politics. Ma'aly et al. [8] explored Deep Learning algorithms for sentiment analysis regarding the Indonesian Presidential Election, demonstrating that neural networks can effectively map voter opinion polarization using online reviews. Furthermore, Prastyapradipta et al. [9] proposed a political hoax detection system for social media, highlighting that combining linguistic feature extraction with Deep Learning effectively detects textual anomalies and manipulative narratives in non-standard Indonesian. To address the complexity of formal Indonesian, Pekandia et al. [10] evaluated IndoBERT [11] on news texts, proving the Transformer's contextual representation superior to CNN-LSTM and Ensemble methods in detecting hoaxes.

While these studies confirm the reliability of NLP in the political domain, none have comprehensively compared various classifier architectures using IndoBERT

representations specifically to resolve extreme data imbalance in political news corpora. To bridge this research gap, this study analyzes the effectiveness of IndoBERT word embeddings by evaluating different classification architectures. Specifically, we compare a Deep Learning group (BiLSTM, GRU, CNN) optimized with Focal Loss against a Machine Learning Ensemble group (XGBoost, Random Forest) utilizing a Frozen IndoBERT approach. A thorough evaluation using the F1-Score metric is conducted to identify the most stable and unbiased model for handling imbalanced political news data..

II. RELATED WORK

Research on sentiment analysis has evolved rapidly alongside the application of advanced computational methods [2]. Initially, text classification heavily relied on traditional statistical methods; however, current trends indicate a significant shift towards computational linguistics modeling [3]. Comparative studies have demonstrated that neural network-based models and Transformer architectures consistently outperform conventional algorithms, such as Naive Bayes, in comprehending sentence context and capturing complex word relations [12], [13].

In Indonesia, the implementation of Natural Language Processing (NLP) in the political and journalistic domains has also been extensively explored. Ma'aly et al. [8] utilized online reviews to analyze public opinion polarization during the Presidential Election, successfully mapping eight sentiment emotion labels using a BiLSTM model that achieved a high performance metric with an AUC of 0.91. In a related context, anomaly detection of information and political hoaxes on social media was successfully conducted by Prastyapradipta et al. [9] using a Deep Learning approach combined with linguistic feature extraction. Furthermore, for handling formally structured news texts, an evaluation by Pekandia et al. [10] proved that a Transformer-based model (IndoBERT) could detect hoax news with an accuracy of 92.24% and an F1-score of 92.23%, significantly outperforming both the hybrid CNN-LSTM model and Ensemble methods.

As the primary foundation for feature extraction in the aforementioned modern studies, the emergence of bidirectional language representation architectures like BERT [14] has revolutionized the NLP field. To bridge the need for understanding Indonesian-specific semantic contexts, IndoBERT was introduced as a pre-trained language model specifically trained on a large-scale text corpus [11]. The reliability of word embeddings generated by IndoBERT has been validated across various text processing tasks, ranging from text summarization for news articles [15] to the detection of online gambling promotions by Liem et al. [16], where IndoBERT proved highly superior, achieving an accuracy of 98.53% and an F1-score of 97.24%, far surpassing the performance of conventional feature extraction methods such as TF-IDF and Word2Vec.

Although IndoBERT is highly robust as a feature extractor, the final performance of text modeling remains heavily dependent on optimization at the classifier layer. Several studies rely on Deep Learning architectures for this stage. The use of recurrent algorithms such as BiLSTM and GRU has proven to be highly robust in Named Entity Recognition

(NER) for Indonesian texts [17] as well as in detecting script-based anomaly attacks [18]. In an emotion recognition study on Indonesian product reviews, Pramana et al. [19] comprehensively compared the BiLSTM model with BERT and Ensemble methods. The results indicated that a standalone BiLSTM with complex preprocessing only achieved an F1-score of 61%, whereas model integration through a weighted stacked generalization (Ensemble) method successfully boosted the peak performance to an F1-score of 75%. This highlights the crucial role of the classifier architecture in capturing local features and sequential context.

As an alternative to Deep Learning, Ensemble Machine Learning algorithms are also widely implemented in classification tasks. Decision tree-based algorithms like Random Forest [20] and gradient boosting algorithms like XGBoost [21] are known for their computational advantages in handling high-dimensional features without being prone to overfitting. The implementation of these Ensemble architectures has proven to be highly adaptive and robust in various cross-domain structured data modeling tasks [22].

Besides the selection of classifier architecture, a major challenge frequently encountered in real-world text classification is the imbalanced data problem [23]. Class distribution imbalance can trigger severe prediction bias, where the model tends to continuously predict the majority class [7]. To address this issue, prior literature has proposed various algorithmic adjustments. One of the most significant innovations is the use of Focal Loss [24], an objective function specifically designed to apply a heavy penalty to easily classified majority class samples. This approach has even been modified and successfully integrated into the XGBoost architecture to resolve centralized imbalanced data classification problems [25].

Based on the literature review above, various classifier architectures ranging from Deep Learning to Ensemble, as well as imbalanced data handling techniques (Focal Loss), have been extensively studied separately across different cases. Nevertheless, no previous research has comprehensively compared the robustness of Deep Learning architectures against a Frozen IndoBERT approach combined with Machine Learning Ensembles, specifically in handling extreme sentiment class imbalances within an Indonesian political news text corpus. Mapping the strengths and weaknesses of each model in this specific domain constitutes the research gap for this study.

III. METHODOLOGY

The experimental steps in this study are comprehensively illustrated in Figure 1. To ensure computational efficiency, particularly during the Deep Learning training phase, the experiments were run on a Virtual Machine (VM) environment supported by 2x NVIDIA Tesla T4 GPUs (32 GB GDDR6 VRAM, 256-bit, 5,120 CUDA cores, 640 Tensor cores, and 640 GB/s memory bandwidth). Overall, this process is divided into four main sequential stages: Data Collection & Preprocessing, Feature Extraction and Imbalanced Data Handling, Classifier Architectures, and Evaluation Metrics, with detailed explanations as follows:

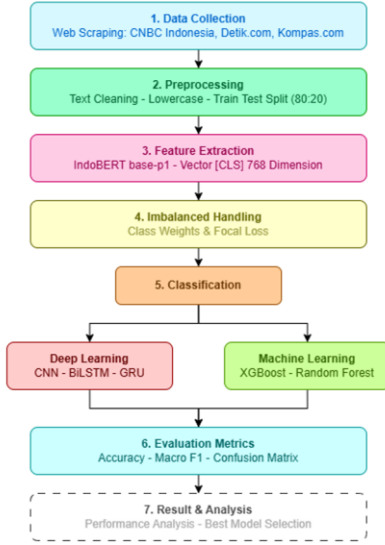


Fig. 1. Research Flowchart

A. Data Collection & Preprocessing

The dataset was acquired through web scraping using the BeautifulSoup library to extract news texts directly from the online portals CNBC (799 data points), Detik (1,000 data points), and Kompas (800 data points). The collected dataset, comprising 2,599 text rows, was then annotated into Positive (411 data points), Negative (574 data points), and Neutral (1,614 data points) classes. The manual labeling process was conducted directly by the research team based on a contextual understanding of political narrative dynamics and journalism in Indonesia, ensuring each text was evaluated objectively according to the direction of public opinion it contained.

In the preprocessing stage, the texts were cleaned of noise (punctuation and special characters) and standardized into lowercase. Stop words were intentionally retained to preserve the complete sentence structure, which is crucial for the attention mechanism in the Transformer architecture [14]. The dataset was then divided into a training set (80%) and a validation set (20%) using a stratified split.

B. Feature Extraction and Imbalanced Data Handling

This study utilized the pre-trained indobenchmark/indobert-base-p1 model for feature extraction, with a maximum sequence length of 512. The selection of the base-p1 (Phase 1) variant was based on the most optimal trade-off between computational efficiency and linguistic representation performance; the large variant risks triggering overfitting on medium-sized datasets and incurs excessively high computational loads, whereas the lite variant sacrifices too many feature dimensions, rendering it less sensitive to complex journalistic language structures [11].

To handle the imbalanced data, a class weight strategy was applied by amplifying the weights of the minority classes (Positive and Negative) to 1.5, while the Neutral class was set to 1.0. In the Deep Learning model group, the loss function was modified using Focal Loss ($\gamma=2.0$) [24] to force the models to focus on hard-to-predict data samples. Meanwhile,

in the Machine Learning group, features were extracted from the 768-dimensional [CLS] token using the Frozen IndoBERT approach [14], which were subsequently normalized using StandardScaler.

C. Classifier Architectures

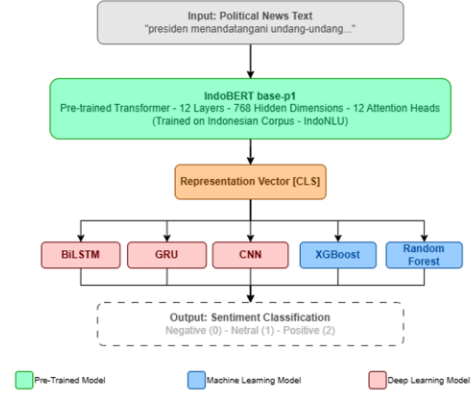


Fig. 2. Sentiment Classification Model Architecture

This study executed two architecture groups. All Deep Learning models were trained for 15 epochs combined with an Early Stopping mechanism to provide sufficient iteration space for the models to learn minority class patterns without triggering overfitting. The learning rate was set to $3e-5$ (with a weight decay of 0.01 and a Cosine Scheduler), which is the most optimal empirical equilibrium value recommended for fine-tuning BERT-based architectures to prevent catastrophic forgetting [14].

- Deep Learning Models:

- IndoBERT Classifier (Fine-Tuning): The pre-trained parameters were fully trained with the addition of a linear layer and a dropout of 0.1 over the [CLS] token representation.
- CNN: Configured using 1D Convolutional layers with 128 filters of sizes 3, 4, and 5, alongside a dropout of 0.2 [13], to capture n-gram local features.
- BiLSTM: Utilized 2 hidden layers with a size of 256 and a dropout of 0.2 [17].
- GRU: Utilized 2 hidden layers with a size of 256 and a dropout of 0.2 [18].

- Machine Learning Ensemble Models:

- Random Forest: Configured using 500 estimators (decision trees), a `min_samples_split` of 5, and a balanced class weight parameter [20] to reduce variance through majority voting predictions.
- XGBoost: Built using 500 estimators, a max depth of 6, a learning rate of 0.1, and a subsample of 0.8 [21] to iteratively correct the residuals (errors) of the preceding decision trees.

D. Evaluation Metrics

Classification performance was evaluated using Accuracy, Macro F1-Score, and Weighted F1-Score metrics. The F1-

Score metric was prioritized as the primary indicator to avoid evaluation bias toward the majority class under imbalanced data conditions [23]. The evaluation phase generated a Confusion Matrix to map the distribution of actual predictions, while the Sastrawi library was implemented separately post-evaluation solely to generate Word Cloud visualizations of keywords per sentiment.

IV. RESULT AND DISCUSSION

A. Overall Model Performance Comparison

This study compares the performance of six classification architectures using IndoBERT embeddings features. The evaluation focused on Accuracy, Macro F1-Score, and Weighted F1-Score metrics. Given the imbalanced condition of the dataset, the Macro F1-Score serves as the most objective indicator because it calculates the average performance of each class without being distorted by the majority sample size.

TABLE I. OVERALL MODEL PERFORMANCE

Model	Accuracy	Macro F1	Weighted F1
IndoBERT	0.8635	0.8257	0.8620
IndoBERT + BiLSTM	0.8635	0.8230	0.8605
IndoBERT + GRU	0.8615	0.8281	0.8606
IndoBERT + CNN	0.8404	0.7909	0.8353
IndoBERT + XGBoost	0.7192	0.5997	0.6991
IndoBERT + Random Forest	0.6981	0.5095	0.6452

Table I summarizes the overall testing results of the models, which are also comparatively visualized in Figure 3. Based on the experimental results, the IndoBERT Classifier (Fine-Tuned) and IndoBERT + BiLSTM models successfully achieved the highest Accuracy of 86.35%. However, when evaluated using the Macro F1-Score metric, the IndoBERT + GRU model emerged as the most stable architecture with a score of 82.81%. Conversely, the Machine Learning Ensembles (XGBoost and Random Forest) recorded the lowest performance.

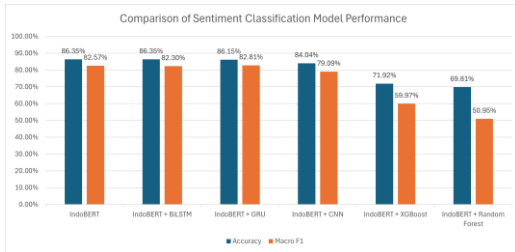


Fig. 3. Comparison of Sentiment Classification Model Performance

B. Performance Analysis by Sentiment Class

To gain a deeper understanding of how each model handles the class imbalance problem, Table II presents the breakdown of Precision, Recall, and F1-Score for each sentiment category (Negative, Neutral, and Positive).

TABLE II. PERFORMANCE METRICS BY SENTIMENT CLASS

Model	Class	Precision	Recall	F1-Score	Support
IndoBERT	Negative	0.8707	0.8783	0.8745	115

	Neutral	0.8879	0.9071	0.8974	323
	Positive	0.7432	0.6707	0.7051	82
IndoBERT + BiLSTM	Negative	0.8293	0.8870	0.8571	115
	Neutral	0.8859	0.9133	0.8994	323
	Positive	0.8125	0.6341	0.7123	82
IndoBERT + GRU	Negative	0.9118	0.8087	0.8571	115
	Neutral	0.8732	0.9164	0.8943	323
	Positive	0.7468	0.7195	0.7329	82
IndoBERT + CNN	Negative	0.7923	0.8957	0.8408	115
	Neutral	0.8653	0.8947	0.8798	323
	Positive	0.8036	0.5488	0.6522	82
IndoBERT + XGBoost	Negative	0.6509	0.6000	0.6244	115
	Neutral	0.7547	0.8762	0.8109	323
	Positive	0.5641	0.2683	0.3636	82
IndoBERT + Random Forest	Negative	0.7313	0.4261	0.5385	115
	Neutral	0.6916	0.9443	0.7984	323
	Positive	0.7500	0.1098	0.1915	82

The Neutral class, which dominates 62% of the total dataset, consistently obtained the highest F1-Scores across all models (ranging from 79% to 89%). The primary challenge lies in classifying the minority classes, particularly the Positive class. The actual prediction distribution of the best model (IndoBERT + GRU) can be seen in the Confusion Matrix heatmap in Figure 4.

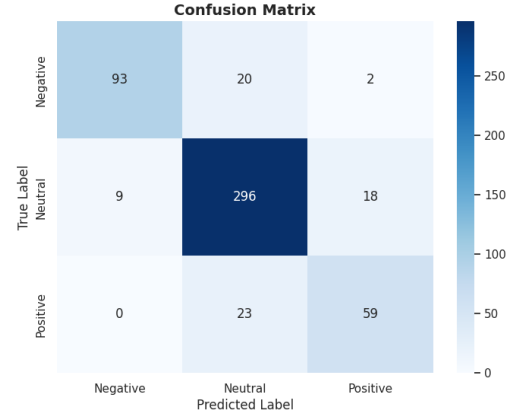


Fig. 4. IndoBERT + GRU Model Confusion Matrix Model

Through Figure 4, it is evident that IndoBERT + GRU proves its robustness in overcoming the imbalance by recording the highest Recall and F1-Score in the Positive class (71.95% and 73.29%, respectively). In contrast, the Machine Learning group exhibited significant difficulty in detecting minority classes. The Random Forest model experienced extreme bias toward the majority class; it achieved a Recall of 94.43% for the Neutral class but failed completely on the Positive class, with Recall plunging to 10.98% and an F1-Score of 19.15%. This phenomenon confirms that the Frozen IndoBERT approach combined with traditional Machine Learning algorithms is not robust enough to resolve the imbalanced data problem in complex political news texts.

C. Qualitative Analysis

In addition to the quantitative evaluation, qualitative analysis was conducted to validate the linguistic context. The extraction of keywords per sentiment is visualized using a Word Cloud in Figure 5, following the removal of stop words using the Sastrawi library.



Fig. 5. Word Cloud Visualization of keyword distribution across Negative, Neutral, and Positive Sentiment Classes

The visualization demonstrates the dominant lexicons successfully separated by the model. The model's success in classifying these categories proves that the proposed architecture can distinguish the emotional nuances and implicit contexts underlying standard journalistic language.

D. Discussion and Practical Implications

An in-depth analysis of the experimental results reveals the fundamental reasons why the Deep Learning group (particularly IndoBERT + GRU) is vastly superior to the Machine Learning Ensembles (Random Forest and XGBoost). The robustness of the GRU lies in its update and reset gate mechanisms, which are capable of remembering the long-term linguistic context of journalistic sentences without losing crucial information. This is combined with dynamic weight updates (fine-tuning) and Focal Loss, which forcefully apply heavy penalties if the model ignores the minority classes. Conversely, the poor performance of decision tree-based algorithms (RF and XGBoost) occurs because node splitting in decision trees tends to focus on global error reduction. Consequently, the high-dimensional Frozen IndoBERT vectors (768 dimensions) were overwhelmed by the 62% dominance of the Neutral class, causing these models to continuously take shortcuts by predicting the majority class to achieve pseudo-accuracy.

If implemented for large-scale deployment, this sentiment model holds significant potential for integration into real-time public opinion monitoring dashboards by governments, survey institutions, or news aggregators. Its implementation requires a data pipeline capable of continuous web scraping (streaming ingestion) and directly forwarding the data to the classification model's API endpoint. With the stability of the Macro F1-Score metric reaching 82.81%, this system can automatically provide early warnings in the event of a sharp spike in negative news sentiment regarding specific political figures or policies, thereby enabling rapid and measurable public opinion mitigation.

Although the proposed model performs highly, this study is not without limitations and obstacles. First, the dataset comprising 2,599 text rows, while sufficient to validate the architecture, remains mid-sized by Deep Learning standards; thus, the model is vulnerable to not recognizing emerging

political vocabulary (slang or euphemisms) in the future. Second, implementing Transformer-based models requires extremely high computational power (such as the use of large VRAM on NVIDIA Tesla T4 GPUs). This makes the deployment costs for real-time inference of millions of news articles expensive. Third, even though the GRU is capable of capturing sequential patterns, the model occasionally still struggles to comprehend high-level language nuances, such as political sarcasm and satire, wrapped in formal journalistic sentences.

V. CONCLUSION AND FUTURE WORK

This study has successfully achieved its objective of evaluating the effectiveness of IndoBERT word embeddings through a comprehensive comparison between Deep Learning architectures and Machine Learning Ensembles for handling extreme data imbalance in Indonesian political news texts. Based on the experimental results, the Deep Learning group, combined with dynamic weight updates (fine-tuning) and Focal Loss, significantly outperformed the Frozen IndoBERT approach applied to decision tree algorithms. The IndoBERT + GRU model emerged as the most optimal and stable architecture, achieving a Macro F1-Score of 82.81%. The robustness of the update and reset gate mechanisms in the GRU proved highly effective in retaining long-term journalistic linguistic contexts while preventing the model from biasing toward the majority class (Neutral). Consequently, it successfully recorded the highest Recall for the minority class (Positive), which had previously been a fatal point of failure for conventional models like Random Forest.

Although the proposed architecture demonstrates highly promising classification performance, there remains ample room for development to overcome the computational and linguistic obstacles encountered during the experiments. For future research, implementing Parameter-Efficient Fine-Tuning (PEFT) techniques, such as LoRA (Low-Rank Adaptation), is recommended to reduce the exceptionally high hardware computational costs without sacrificing model performance. Furthermore, future studies are highly encouraged to expand the dataset corpus volume and explore Aspect-Based Sentiment Analysis (ABSA) approaches. This would enable the model to not only predict sentiment globally but also precisely identify the specific political entities or figures targeted by the sentiment, as well as decipher complex linguistic nuances, such as sarcasm and satire, within formal journalistic texts.

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